

ITS67204 Professional Practices & Information Security

Group Assignment (30%)

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1.0 Introduction

This report is a multi-dimensional analysis of the Uber data breach and cover-up of 2016, which is a landmark case because it captures the multifaceted nature of interactions between technology, ethics, law, and corporate governance in the digital era. This incident resulted in the theft of the personal information of 57 million drivers and customers such as their name, phone number, email addresses, and details on their driver license. With its far deeper meaning, it is the decision of the corporation that followed to cover up the breach in a systematic manner (Greenberg, 2017).

This report analyses the Uber breach using five dimensions which are intertwined. The technical aspect looks into the vulnerabilities of the GitHub repository and AWS infrastructures of Uber; it evaluates whether the failures were due to ineffective protection, human factor, or negligence of the system. The professional and moral aspect determines the void of the basic responsibilities made by IT professionals and the leadership, analyzing the case with the help of deontological responsibilities to the transparency, utilitarian actions to minimize harm, and ethical issues of professional integrity in terms of virtue ethics. The legal and regulatory aspect looks at the case against the frameworks such as the GDPR and U.S. breach notification laws, which are found to have loopholes in their cross-jurisdictional application. The analysis of organizational and societal impact tracks the way that the cover-up caused a loss of trust among the population, disrupting the trust of the investors, and prompting even a greater concern regarding the stewardship of corporate data. Lastly, the futuristic policy suggests that a unified prevention system be implemented which would merge both technical measures, ethical governance and regulatory changes as well as cross-sector co-operation.

As this case demonstrates, effective data protection cannot be founded on technical defenses only, as it has to be supported by sound ethical governance and disclosure. Companies should combine both technical protection with well-developed ethical practices that propagate a sense of responsibility, create secure whistleblowing systems, and implement uniformity in disclosures across the world. By doing so, companies will be able to reduce the risk of cybersecurity, as well as show their seriousness about being responsible data stewards, thus being able to earn the goodwill of the long-term of conducting business in the current digital environment.

2.0 Case Background

Back in 2016, Uber Technologies Inc. was one of the fastest-growing ride-hailing platforms in the world. However, the company suffered a data breach that revealed the personal details of 57 million customers and drivers worldwide (Lee, 2017). Apart from showing that the company's cybersecurity measures were inadequate, the incident highlighted major problems with Uber's culture and governance. At the time, the company was worth more than US\$60 billion and was engaged in aggressive global expansion. However, its internal processes and ethical practices were far from keeping pace with its rapid technological growth (Isaak & Hanna, 2018).

First, two hackers gained unauthorized access to Uber's private GitHub repository. They found the code necessary to access the company's Amazon Web Services (AWS) account. With these codes, the hackers obtained all the sensitive user and driver data, including names, email addresses, phone numbers, and over 600,000 driver's license numbers (*Uber Settles FTC Allegations That It Made Deceptive Privacy and Data Security Claims*, 2025). Rather than informing the authorities and those affected, Uber gave the hackers \$100,000 as if it were a reward for their "bug bounty" program (Greenberg, 2017). This measure exemplified the "win-at-all-costs" mentality still prevailing at Uber, where restoring the company's image was prioritized over transparency and accountability.

The longer Uber waited to reveal the breach, the more harmful it was to their public image. When the incident came to light in November 2017, it was very close to Travis Kalanick handing over the CEO position to Dara Khosrowshahi. Khosrowshahi did well by acknowledging that the breach had been concealed and that it was an intolerable infringement on public trust. He also made it clear that he was ushering in a new era of corporate integrity (Wong, 2017). In response, regulators in many countries launched investigations. As a result, Uber finally paid \$148 million in settlements with U.S. state attorneys general, which was the largest data breach settlement at the time (*Uber Settles FTC Allegations That It Made Deceptive Privacy and Data Security Claims*, 2025).

The impact of the Uber breach was not limited to finances, it was a disaster for the company. During an important phase of its worldwide expansion, Uber lost its position as a technology

leader and the trust of its customers. The public branded the company as an ethical crisis of Silicon Valley, where innovation is often accompanied by a lack of moral responsibility (Isaak & Hanna, 2018). Researchers have pointed out that, in Uber's case, the data breach exposed not only the company's inability to govern its technology, but also the technology itself. This is especially true when security and corporate non-compliance priorities are given precedence over adherence to ethical standards (Martin et al., 2017). Uber's decision to hide the breach was not merely a technical mistake, but rather a symptom of a lack of transparency within the organization.

Strategically, the crisis was a pivotal moment for the company's leadership to reform. Khosrowshahi spearheaded Uber's trust-building approach and internal reforms. Steps such as enhancing cybersecurity infrastructure, appointing a Chief Privacy Officer, and strengthening data governance policies were taken (Wong, 2017). The company also increased engagement with regulators and introduced mandatory data ethics training for employees. These actions signaled a shift toward ethical digital governance, emphasizing that long-term success in the digital economy depends on protecting stakeholder trust as much as technological innovation (Greenberg, 2017).

The 2016 Uber data breach remains a strong warning to digital platform firms engaged in data collection. The incident showed that security breaches are not just technical failures, but rather reflections of an organization's core values. Uber's saga shows that corporate accountability and digital trust must develop together and be interdependent. If a company misuses transparency, the consequences will be more than paying fines, and the brand may lose its reputation and customers' loyalty forever (Sydney, n.d., 2021).

The Uber situation reminds both academia and the business world that, in a data-driven environment, being ethical and accountable is more important than coming up with new ideas. Companies that do not embrace these values will lose the trust that is the foundation of their prosperity.

3.0 Technical Analysis of the Breach

3.1 Attack Vector and Exploited Vulnerabilities

The 2016 Uber data breach is a powerful case for multidimensional analysis of Global Cybercrime, Ethics and Security. It shows how inadequate security measures, poor management and human error will lead to large scale leaks of information. The attack was not achieved through strong technical skills, but through weak internal controls and security lapses. Approximately 57 million users worldwide were affected, including 60 thousand drivers (Greenberg, 2017).

The fundamental vulnerability in this data breach lies in the failure to properly monitor employee access to consumer and driver data and the failure to maintain reasonable security measures. One of the Uber engineers mistakenly left credentials in the GitHub repository, violating basic secure development practices (Whittaker, 2018). This allowed an attacker to access Uber's source code on GitHub, find a cloud credential, and use it to download Uber user's data (Lydia, 2019). The attacker downloaded a backup of sensitive data containing names, email addresses, phone numbers, and driver's license information (Balakrishnan & Bosa, 2017).

This incident revealed weaknesses in Uber's cloud security architecture, especially in areas such as Identity and Access Management (IAM) and Real-Time Intrusion Detection. Moreover, Uber's not to implement basic repository scanning tools, this shows a clear lack of preventive measures. The overall attack vector reflected not just a single technical oversight; it powerfully shows a systemic failure in enforcing cybersecurity hygiene and awareness across the engineering team.

3.2 Nature and Classification of the Attack

This attack was an independent act by hackers for financial gain, not state sponsored. After successfully getting Uber users data, they require a six-digit amount in exchange for the deletion of the stolen data. A former Uber executive used the company's bug bounty program to pay 2

hackers 100 thousand. This is to delete the data they had stolen without informing users or authorities to conceal the data breach (Whittaker, 2018).

According to the Federal Trade Commission (2017), Uber was accused of not taking affordable and reasonable steps to prevent the data breach. The Commission emphasized that no advanced hacking methods or malware deployment were needed. This attack was technically straightforward and avoidable. Instead, this incident used human negligence and improper safety supervision, thus constituting a data breach rather than a purely technical one (Federal Trade Commission, 2017). This data breach incident can be classified as human factor driven, because the key weaknesses lay in internal behaviour, poor access management and absent monitoring, rather than in the sophistication of the attacker tools.

3.3 Technical Safeguard Failures

Uber's security vulnerabilities can be summarized into several key technical flaws, including:

- 1. Credential leakage:** Uber did not require engineers and programmers to use different access keys to access personal information stored in the cloud (Lydia, 2019).
- 2. Lack of access monitoring:** Uber rarely monitored internal access to user and driver personal information, resulting in its security measures failing to provide adequate safeguards (Lydia, 2019).
- 3. Inadequate encryption measures:** Uber stored sensitive consumer information in plain text in a cloud-based database backup (Lydia, 2019).
- 4. Multi-factor authentication (MFA) not enabled:** Social engineering was the root cause of the Uber data breach, and the fatigue of MFA was the root cause of the hacking (Salmon, 2023).

These shortcomings show that Uber's cybersecurity approach is reactive rather than proactive. They always emphasize correction over prevention of data breach. This attack could have been easily prevented or mitigated if the correct security measures had been taken earlier.

3.4 Technical Assessment and Root Cause Evaluation

According to Salmon (2023), Uber's enhanced Privileged Identity Management and Privileged Access Management (PIM/PAM) solutions enable organizations to control, monitor, and protect privileged identities and access to critical systems. Maintaining detailed audit trails helps organizations track privileged access and detect anomalous activity, enabling rapid responses to potential breaches (Salmon, 2023).

This data breach also shows a deeper oversight by Uber in terms of security governance and organizational culture. At the time, Uber did not require the review of code submitted to the code repository, which allowed sensitive credentials to be uploaded without peer verification. Uber also lacked automated credential-scanning tools (GitGuardian or similar secret-detection systems), which would have identified exposed keys before attackers could exploit them. Uber also failed to clearly separate development and production environments, enabling developers to access live systems with broad and unsupervised privileges. These security culture weaknesses indicate an institutional failure rather than an isolated case of individual human error.

The three major factors of this data breach include: Human error (Improper handling of credentials and lack of security awareness training), organizational negligence (Failure to effectively implement security strategies and audits) and inadequate technical oversight (lack of basic security configurations and monitoring tools).

The main key to root cause assessment is understanding why these vulnerabilities initially existed. According to Bhuiyan & Milmo (2022), we can determine that Uber's rapid global expansion forced it to prioritize speed, growth, and product deployment, while overlooked basic security investments (Bhuiyan & Milmo, 2022). This situation lets Uber security programs, internal audits, and governance structures not scale at the same pace as the company's technological development. This cause imbalance contributed to insufficient oversight, outdated internal policies, and underfunded security practices. As a result, we can know fundamental security controls such as code reviews, credential rotation, and monitoring were overlooked, creating long-term structural gaps that enabled the breach.

In conclusion, the attack on advanced hacking methods or malware deployment were needed, this was technically straightforward and avoidable. This incident used human negligence and improper safety supervision, thus constituting a data breach rather than a purely technical one (Federal Trade Commission, 2017).

4.0 Professional Ethics Violations

4.1 An analytical framework for systemic ethical misconduct

The 2016 Uber data breach case represents a profound failure in professional ethics that extends far beyond the technical incident itself. The core ethical transgression lies not in the initial system compromise but in the deliberate and systematic cover-up orchestrated by professionals in positions of significant trust and responsibility. This section provides a comprehensive analysis of these ethical failures, examining the breach of fundamental duties, the specific failings of key actors, the corruption of professional norms, and the enabling role of organizational culture (Mandarano, 2023).

4.2 Breach of Fiduciary Duty and the Public Trust

The most critical ethics violation is the violation of fiduciary duty. With the responsibility of data protection, CSO Joe Sullivan and his team members chose to give a ransom to hide the truth at the expense of user privacy and security and prioritizing a short-term reputation (United States Attorney's Office, Northern District of California, 2022).

The actions that Sullivan took turned out to be active conspiracy due to a conflict of interest in relation to an ongoing FTC investigation. He settled on valuation over safety to the populace to prevent the risk of endangering settlements (Former Chief Security Officer of Uber Convicted of Federal Charges For, 2022). His act of masquerading extortion as a bug bounty was an attempt to show intention to lie considering that the lasting damage of lying is much greater than the actual act (Newman, 2022).

Eventually, this deception denied 57 million users independence and an informed decision and could not take any precautions themselves and instead reduced them to instruments of company stability (Greenberg, 2017).

Moreover, by writing NDAs that lied about the ransom as a bug bounty, the in-house counsel was also complicit within the company, using the legal tools to mask a felony, which is known as institutional complicity (K.Dwyer, 2022). Other professionals had not performed their oversight functions, and the team members had put organizational loyalty first before the duty of the citizens. This is an example of the fallacy of following orders. The sentence is a harsh reminder to the heads of security that no one should ever defend the company by violating the law.

4.3 Corrosion of Industry Standards and Organizational Culture

The ethical misdeeds of Uber significantly played off its bug bounty program by turning a loyalty system into a vehicle of hush money in order to cover up criminal activities. Uber made the difference between cooperation and criminality gray by redefining extortion as research illegally, which affected the international integrity of the vulnerability disclosure programs. Such secrecy only worsened the distrust that people had of the cybersecurity career. To avoid such cases in the future, industry organizations need to create ethical standards that consider concealment as a misconduct and optimize transparency to make sure that bug bounty programs do not whitewash any illicit activity (HackerOne, 2019).

This institutional malpractice is a result of the Uber organizational culture whereby a win-at-any-cost culture encouraged regulatory avoidance and saw ethical lines as a hindrance. There were no effective internal controls at all, no independent compliance functions or whistleblower protection to come up with restrictions. This laxity system enabled unethical decisions to pass unhindered.

Therefore, companies must commit to building a “safety culture” that encompasses not only technical safety but also psychological and ethical safety. This requires structural reforms such as appointing a Chief Compliance Officer reporting directly to the board, establishing independent and protected internal reporting channels, and integrating ethical compliance performance into executive evaluations. Only by making ethics an immovable cornerstone of the organization can

decisions sacrificing long-term credibility for short-term gains be fundamentally eliminated (Cook, Patel, & O'Rourke, 2018).

4.4 Ethics Reflection

In summary, the professional ethics violations in the 2016 Uber data breach were severe, systemic, and multi-faceted. They involved a conscious and active choice by credentialed professionals to deceive the public, regulators, and their own corporate oversight bodies. This case was not a simple error in judgment but a fundamental abandonment of core ethical duties, including fiduciary responsibility, honesty, and transparency. It stands as a stark reminder that technical expertise is insufficient without being grounded in unwavering ethical principles, and that without a supportive and accountable ethical culture, even the most skilled professionals can catastrophically fail in their most important duty: to protect the public trust.

5.0 Application of Ethical Theories

5.1 Ethical Evaluation Framework

The ethics of Uber's 2016 breach can be considered more deeply through the lens of moral philosophy frameworks. We will go on to discuss in more detail why these were unethical acts.

Using the three theoretical approaches – deontology, utilitarianism, and virtue ethics – in the discussion reveals in more detail the nature of these moral transgressions and how they also indicate professional irresponsibility in the field of cybersecurity.

5.2 Duty-Based Ethics

Deontological ethics focuses on rules and duties to determine what is morally right or wrong, rather than the results of those actions, and in considering whether it is morally right or wrong it asks, does this action conform to my duties or the requirements of a rule? And the obligation to “do the right thing” is just that, more important than the consequences, Rueter (2025) said.

In the case of Uber, its leaders had a clear moral and professional obligation not only to notify the breach and protect user information, but also to be transparent about doing so, with a special focus on truthfulness, accountability, and the ACM Code of Ethics (2020) highlighting that sharing these values they had should be with them. But the company chose to cover it up — and paid off hackers to the tune of \$100,000 to delete the stolen data. The action went so blatantly against the duty of transparency and honesty to which a professional is bound. While the administration ascribed the intent to spare worrying and protecting corporate image, deontological perspective would oppose this thinking as one cannot place one's moral duties above the dis-comfort or one's own benefits (Rueter, 2025).

And finally, Uber's silence about the breach was itself unethical, further reflecting an absence of commitment to the company's obligations to customers, employees, and regulators.

5.3 Utilitarianism Ethics

Utilitarianism takes moral decisions to be those that promote the greatest good for the greatest number, maximizing overall happiness and reducing suffering. According to Rueter (2023), this moral framework is concerned with the maxim that is best for the majority, with the view that right actions are those for the overall good, not the individual gain, of society.

From a utilitarian perspective, Uber managers may believe that keeping the 2016 hack under wraps would prevent user panic and short-term financial losses. However, utilitarianism is not about end products, but rather overall long-term consequences of actions. Mill (1863/2001) also emphasized that moral action should create durable public benefit, not merely prevent transient inconvenience. Keeping the truth under wraps, Uber actually did more harm to users, employees, and business reputation.

When the truth finally came to light, the damage inflicted far outweighed the fleeting stability gained by concealment. User personal information remained vulnerable to misuse, employees felt deceived, and public trust plummeted. These consequences clearly demonstrate that Uber's decision did not yield an overall positive impact.

As Shaw (2016) elaborates in business ethics research: Open and responsible corporate conduct often generates higher ultimate total benefits for organizations and other stakeholders. Had Uber addressed the incident with candor, it could have avoided widespread harm and built lasting trust. Even from a utilitarian perspective, Uber's approach is difficult to justify—the harm inflicted far outweighed any actual benefit.

5.4 Virtue Ethics

Virtue ethics centers on examining the moral character and intent behind one's actions as opposed to looking at the actions themselves. What does a virtuous organization look like within an audience of risk or public pressure? Rueter (2024) Further, virtue ethics highlights one's moral progress as a process of continually embodying virtuous actions and reflecting upon one's lived experience, that is, true morality springs from being of good character and not rule-following per se.

In Uber's 2016 data breach, the leadership's hiding of the incident showed a failure of integrity and courage. Displaying responsibility, not shamefacedly yet still scrambling to protect its image and not its moral accountability. This is consistent with Aristotle's (2018) claim that virtuous action is dependent upon virtuous disposition; without moral courage, an organization will fail to act properly in the face of opposition.

With respect to business ethics, Hartman et al. The cover-up by Uber, therefore, revealed a profound ethical flaw in its leadership culture. Transparency and empathy in response to those who are compromised is how a virtuous organization would respond, reminding us that moral awesomeness is just as critical in cybersecurity governance as technical competence.

5.5 Overall Ethical Reflection

Each ethical theory looks at Uber's actions from a different angle, but they all come to the same conclusion.

Ethical Theory	What It Focuses On	What It Says About Uber
Deontology	Doing your duty and following moral rules	Uber failed to be honest and responsible
Utilitarianism	Looking at overall results and public good	Uber's choice caused more harm than good
Virtue Ethics	Showing good character and integrity	Uber's leaders lacked honesty and courage

While these theories pertain to different subjects, they all concur that Uber's choice to conceal the breach was indefensible. "It wasn't just one misstep—it was indicative of a deeper moral failing in the leadership and culture of the company."

5.6 Key Takeaways from the Uber Case

The Uber case illustrates that cybersecurity breaches that exploit technical vulnerabilities are also enabled by the absence of treat ethics governance.

Organizations should address three key areas to stop this from happening again.

For one, they must develop clear, mandatory data breach notification rules to provide visibility and accountability.

Second by reinforcing their internal ethical culture, through the establishment of whistleblower protection mechanisms and communications channels conducive to the responsible notification of concerns.

And continuous ethics and compliance training for executives and their cybersecurity teams can help sustain morally-minded day-to-day decision making.

By applying ethical standards in the practice of security management itself, organizations can regain the trust and preserve it over time. The bottom line is that technical safety without moral safety is still vulnerable.

6.0 Legal and Regulatory Implications

6.1 Overview of the Legal Fallout

The Uber incident of 2016 escalated into a significant legal case due to how the company handled the breach. Nevertheless, the significance of the Uber breach lies in how the response to the breach led to severe repercussions for the company. However, the lawsuit in the Uber breach was not directed at the initial failure of Uber's cybersecurity measures (de la Torre, 2019; United States Department of Justice, 2022).

6.2 Criminal Liability of the Former Chief Security Officer

Chief Security Officer Joe Sullivan was convicted for obstruction of proceedings of the Federal Trade Commission and misprision of a felony. Sullivan paid \$100,000 to attackers through the bug bounty program for Uber, misrepresented nondisclosure contracts that attackers did not take any data, and also refused to disclose the information to the FTC during an ongoing investigation (United States Department of Justice, 2022; de la Torre, 2019). This was the first time a corporate executive was criminally convicted for concealing a data breach rather than for weak security controls (Dwyer, 2022).

6.3 Emerging Duty to Disclose and Regulator Expectations

In the Sullivan trial, an implicit obligation to report further incidents exists once the relevant authority is scrutinizing the data handling practices of the concerned firm (Dwyer, 2022). At the time of the 2016 data breach, Uber was already being probed by the FTC. Keeping the breach in-house was interpreted as active concealment rather than a simple delay in notification (United

States Department of Justice, 2022; de la Torre, 2019). Thus, the importance of explicit communication lies in avoiding the implications of concealment.

6.4 Assessment Under International and National Regulatory Frameworks

The incident indicates the presence of many shortcomings in the world's data protection regulations. According to GDPR regulation (applicable from 2018), Uber's one-year concealment will breach GDPR's Articles 33 and 34. Penalties of maximum €20 million or 4% of the world-wide business will follow. This clearly surpasses the \$148 million agreement in the United States (European Parliament & Council of the European Union, 2016). CCPA (applicable from 2020) will follow punitive damages of \$750 per individual. The NIST Cybersecurity Framework shows failures across all five key functions: inefficient asset management (Identify), insecure access controls (Protect), inefficient detection (Detect), response strategy centered on concealment (Respond), inefficient recovery plans (National Institute of Standards & Technology, 2018). The Budapest Convention exposed practical resource limitations in prosecuting cross-border attackers. (United Nations Office on Drugs & Crime, 2001).

HIPAA applies only to healthcare entities, so Uber falls outside its scope. The EU AI Act (2024) targets risks related to AI systems, and is not relevant to this case. Nevertheless, it was not applicable in the course of the classic failure of security. Several regulatory gaps contributed to both the breach and its concealment: the patchwork system of state laws in the U.S. left regulation ambiguous along the lines of compliance; the weakness in the regulation of executive accountability was overridden in favor of reputation over duty; the absence of international harmonization made way for the problem of regulation arbitrage; and the regulation on bug bounties made it simple to mistake the payments for blackmail (De la Torre, 2019; Dwyer

6.5 Cross-Jurisdictional Comparison: United States vs. European Union

Also, the U.S. lacked comprehensive federal privacy laws. Litigation concerned the violation of an existing FTC consent decree and the willful attempt to keep the breach secret rather than the breach per se. Sullivan's lawsuit was for obstruction of justice and misprision of felony for the

attempt to conceal. The \$148 million settlement was negotiated by the various state attorneys general, indicative of decentralized enforcement (De La Torre, 2019; United States Department of Justice, 2022).

Under the GDPR regulation, the breach would require obligatory notification within 72 hours (Articles 33-34), with fines not exceeding 20 million euros or 4 percent of the world-wide business turnover. The EU adopts harmonized enforcement by supervisory bodies coordinated by the European Data Protection Board (European Parliament and Council of the European Union, 2016).

There are also three major points of departure. Firstly, the U.S. system was based on scattered state laws enforced by the FTC, in contrast to the EU's 72-hour notice system. Secondly, in the U.S. system affectation of crime was prosecuted through existing terms (obstruction of justice/misprision), while the GDPR system outlines dedicated fines for failure. Notably, the GDPR system imposes penalties based on revenue. Thirdly, the U.S. system was the first to charge an individual officer criminally. These elements demonstrate the disparity in the two regimes in terms of the absence of uniform standards, meaning companies exploit loopholes (Dwyer, 2022).

6.6 Implications for Incident Response Governance and Digital Evidence

Evidence in the criminal trial was mainly based on emails within the institution, payments, and nondisclosure agreements that clearly indicated the intention to conceal the breach (United States Department of Justice, 2022; de la Torre, 2019). Documentation during the process of managing the crisis was used by the courts to indicate concealment rather than clarity. Organisations should bring all aspects of the law into the process of responding to the crisis through clear notification processes, documentation of payments in clear terms, and appropriate communication processes (Dwyer, 2022). Poorly managed incident response can create criminal liability for organisations and decision-makers because of the importance of notification in the business's governance aspects rather than in public relations (de la Torre, 2019).

7.0 Organizational and Social Impact

The Uber data breach in 2016 was not only a technical glitch, but also exposed deeper problems in the company's organizational structure and responsibility culture. Subsequent developments have shown that cyber security incidents not only cause damage at the system level, but also affect the operation mode within enterprises and the public's trust attitude towards digital platforms.

7.1 Organizational-level impact

7.1.1 Reputational damage

Uber's decision to conceal the data breach has had a long-term impact on its reputation. Companies once regarded as symbols of innovation have suddenly become typical examples of ethical issues in the technology industry (Isaak & Hanna, 2018). After the incident was exposed, stakeholders and users began to doubt whether the values promoted by Uber were true and reliable (Lee, 2017), and the lost trust could not be re-established in a short time.

7.1.2 Leadership transition and cultural adjustment

After the incident, there were also changes in the company's senior management. The departure of the former CEO Travis Kalanick and the appointment of Dara Khosrowshahi mark Uber's attempt to transform towards a more transparent and compilation-oriented direction (Wong, 2017). The company subsequently established the position of Chief Privacy Officer and strengthened privacy and ethics training. Although these measures cannot immediately solve all problems, they at least demonstrate the willingness to attempt to rebuild internal values.

7.1.3 Financial and operational impacts

From a financial perspective, Uber paid a huge price for this, including a \$148 million settlement to a U.S. state attorney general (the largest data breach settlement at the time) (Uber Settles FTC... (2025) In addition, the company also bears a large amount of legal and compliance costs, which prevent resources from being invested in innovation or operational improvement. Investor

confidence has also been hit, and Uber can only restore stability by strengthening internal governance and risk control.

7.1.4 Employee Morale and Organizational Trust

Within the company, many employees are uneasy about the actions of the top management. The ethical decisions made by the leadership can directly affect morale, and some employees have begun to question whether the values emphasized by the organization are still credible. Therefore, restoring internal trust has become as important as enhancing technical security measures.

7.1.5 Changes in Strategy and Governance Structure

In the long run, this incident has prompted Uber to reconsider its priorities. The company is gradually shifting from its previous model of "focusing on rapid expansion" to a path that prioritizes transparency and trust as more important factors. Senior management is considering the issue of moral responsibility. Although this is a slow process, it is still a strong signal that Uber is trying to learn from its past mistakes.

7.2 Social Impact

7.2.1 Public trust in digital platforms has declined

The incident has deepened people's concerns about the way technology companies handle personal data. The public was already uneasy about data collection and use, and this incident has further intensified this suspicion, posing an even greater trust crisis for digital services that rely

on sensitive data.

7.2.2 Promote regulatory reform

This incident also affected the response of global regulatory authorities. Many legislators take Uber as an example, arguing that the current regulations are insufficient and that stricter requirements need to be introduced, such as the GDPR and the updated leak reporting regulations in the United States (Greenberg, 2017). Transparent and timely notifications have

gradually evolved from "suggestions" to the basic expectations of the public and regulatory authorities.

7.2.3 Legal precedents for management accountability

The conviction of Joe Sullivan, the former chief safety officer of Uber, has become a significant legal turning point. For the first time, a senior executive has been held personally criminally responsible for concealing data breaches, sending a clear signal that in cybersecurity governance, intentional concealment or misguidance may lead to personal consequences, not just the responsibility of the company.

7.2.4 Impact on the Cybersecurity and Ethical Hacker Community

Uber's approach to managing hacker payments has sparked intense debate in the field of cybersecurity. The approach through bug bounty programs is regarded as crossing the line between responsible disclosure and similar extortion. Many researchers believe that vulnerability reporting channels are based on honesty and collaboration. This case has also drawn increasing attention from the industry to issues of ethical standards and transparency mechanisms.

7.2.5 The Impact of Media and Ethical Hacker Communities

This incident was exposed through an investigation by Bloomberg rather than being disclosed voluntarily by Uber, which shows that the media can play an external supervisory role when enterprises lack transparency. The relevant reports not only affected the public's understanding of the incident but also pushed Uber to make a more serious response. Meanwhile, the ethical hacker community has also expressed concerns, believing that this incident may undermine the trust foundation of the vulnerability reporting system. The reactions of the media and the hacker community jointly shaped society's perception of this incident and also influenced the public's expectations of how enterprises should handle security incidents.

7.3 Overall Reflection

In short, the Uber data breach in 2016 demonstrated that the consequences of cybersecurity issues even affected individuals, organizations and regulatory levels. It can influence the essence of an organization, the level of public trust and the direction of regulation. This case highlights the fact that it has been proven that making public data processing practices and taking on ethical responsibilities have become the core foundation for the survival of digital enterprises. Once trust is lost, it is very difficult to restore. This requires enterprises to constantly demonstrate a responsible corporate image and stable data management practices at all levels.

8.0 Preventive Strategies & Future Framework

The Uber breach in 2016 really shows that cybersecurity problems often don't come from just one place. It wasn't only about a weak password or a leaked key. It felt more like a mix of several internal issues—unclear responsibilities, poor decisions, and a culture that didn't treat security seriously. So, preventing similar incidents requires more than “fixing the tech.” It needs changes in leadership behaviour, legal compliance, and even how different teams work together. Uber did try to repair things afterwards, such as hiring a Chief Privacy Officer and improving its data governance process (Khosrowshahi, 2017). These changes at least prove that companies can improve if they actually commit to it.

8.1 Technical and Operational Safeguards

The breach happened because of a few basic but very common mistakes: exposed GitHub credentials, weak access control, and unencrypted cloud data (Lee, 2017). These are the kind of issues that most companies know they should fix but sometimes don't prioritise. Using SIEM tools for monitoring, applying Multi-Factor Authentication (MFA), and having proper Identity and Access Management (IAM) can already prevent a lot of trouble (Furneaux, 2016). Encrypting important data, whether stored or being transferred, is also something that should have been done earlier (Khosrowshahi, 2017).

Human error continues to be a huge challenge. Even with good tools, one careless click can undo everything. So regular training—phishing drills, teaching staff how to handle credentials safely—matters more than many people think (Salmon, 2023). Adopting Zero Trust Architecture (ZTA) helps reduce internal risk because it checks every access request instead of assuming trusted users (National Institute of Standards and Technology, 2020). Some companies also use honeypots, red-team practices, and immutable backups as extra layers of protection.

8.2 Ethical and Governance Culture

What made the Uber case worse was the decision to hide the breach. Calling the ransom a “bug bounty” reward was widely seen as unethical (Wong, 2017). This shows that strong security tools won’t help if leaders choose the wrong approach during a crisis. Ethical guidelines like the IEEE Code of Ethics (IEEE, 2020) and ACM Code of Ethics (ACM, 2018) exist for situations exactly like this.

During difficult moments, transparency is more important than short-term reputation management. Staff also need a safe space to raise concerns without fear. Boards should consider setting up independent committees to look at ethical and security matters. Once ethical behaviour becomes normal in daily work, cybersecurity won’t rely only on technical teams—it becomes part of the organisation’s values.

8.3 Legal and Regulatory Compliance

After the 2016 breach, Uber ended up paying about \$148 million, and the Chief Security Officer even faced criminal charges for the way the incident was handled (Federal Trade Commission, 2018; Department of Justice, 2022). So hiding the breach clearly made things worse. It basically shows that not reporting a security incident on time can cause more legal trouble than just being open about it from the start.

Because of that, companies should have simple and clear steps for what to do when a breach happens. This includes who to inform, how fast to report, and how communication is supposed to work. These steps also have to match laws like GDPR, CCPA, and Malaysia’s PDPA (European

Parliament & Council, 2016). A lot of organisations think they follow these laws, but when something happens, they realise their procedures are old or not detailed enough.

Another problem is third-party services. Many companies use cloud vendors, but their contracts don't always say what the vendor must do during a breach. If these responsibilities aren't written clearly—like reporting time or security duties—the company might still get blamed even if the vendor caused the issue. So legal compliance is not just about following rules; it's about having updated procedures and making sure every external partner follows them too.

8.4 Cross-Sector Collaboration and Continuous Improvement

Cybersecurity issues spread faster than any single organisation can handle alone. Because of this, information-sharing networks like ISACs are helpful. Companies in these groups exchange threat intelligence and practical solutions (Lohrmann, 2017). Standards such as ISO 27001 and the NIST Cybersecurity Framework also help maintain a consistent approach.

Incident-response simulations that involve IT teams, legal advisors, management, and communication staff are also useful. These drills reveal weaknesses that might not appear in normal operations. Tracking indicators like mean time to detect (MTTD) and mean time to respond (MTTR) helps organisations understand whether improvements are real or just theoretical.

8.5 Integrated Future Framework

A long-term cybersecurity strategy usually comes from four connected elements:

- Ethical leadership — leaders choosing honesty and responsibility
- Technical protections — the tools and systems that prevent attacks
- Regulatory compliance — following laws and reporting requirements
- Cross-sector cooperation — working with others, not in isolation

When these elements support each other, organisations can build a more complete cycle of prevention, detection, response, and recovery.

8.6 Implementation Roadmap and Prioritisation

In the first six months, organisations should focus on basics such as enforcing MFA, reviewing cloud security settings, and updating breach-notification procedures. Leadership must also make their stance on ethical conduct very clear (Salmon, 2023).

During the 6–18 month period, companies can strengthen their structures. This includes creating an ethics and compliance committee, holding more regular penetration tests, and offering more advanced training to staff.

After 18 months, the focus shifts to making these practices permanent. Publishing transparency reports, inviting external auditors, and participating in global cybersecurity alliances help maintain long-term consistency. Tracking indicators like ethical-awareness scores and response times is useful for evaluating progress.

8.7 Global Applicability & Adaptability

A cybersecurity model should work for organisations of different sizes and in different legal environments. Basic measures like MFA, encryption, and staff training are affordable for most companies. GDPR can serve as a useful baseline for cross-border consistency and matches well with regulations like CCPA, PDPA, and LGPD (European Parliament & Council, 2016).

The Uber case also demonstrates how prevention is cheaper than recovery. The cost of implementing basic safeguards was less than 1% of Uber's overall losses, which went over \$2.15 billion (Federal Trade Commission, 2018). This shows that strong data protection is not only a legal need but also a long-term advantage (Martin et al., 2016).

9.0 Conclusion

The data breach and cover-up incident of Uber in 2016 clearly demonstrated that the security crisis in the digital age is never a single technical issue, but a systemic risk caused by the failure of multiple aspects such as technology, ethics, law and organizational culture. This case offers three key insights for all digital enterprises.

First of all, prevention is always more valuable than remedy. Uber has paid a huge compensation and suffered continuous damage to its reputation for covering up the incident, while the cost of establishing a sound safety governance system is only a tiny fraction of the cost of the accident - the analysis in this report shows that the preventive investment is less than 1% of the actual loss. This proves that regarding cyber security as a burden rather than a strategic investment is an expensive and dangerous misjudgment by enterprises.

Secondly, trust has become the most scarce and difficult-to-restore asset in the digital economy. Technology can be fixed and systems can be rebuilt, but once public trust is broken, the cost of repair far exceeds the expectations of enterprises. As the regulatory environment becomes increasingly strict, enterprises can no longer rely on legal gray areas, taking chances or apologizing afterwards to recover their losses. Ethical governance must shift from being "optional" to a fundamental element concerning the survival of enterprises.

Ultimately, true accountability requires multi-level support. When internal governance is inadequate, media supervision, whistleblower actions and regulatory enforcement often become the last line of defense. However, relying on external forces to correct errors is not only lagging behind but also costly, and it also means that the organization can no longer independently maintain its governance system. Resilient enterprises must internalize transparency, responsibility and ethics into their organizational culture, making the defense line an inherent ability of the enterprise rather than a passive response after a crisis.

Although Uber has demonstrated the possibility of organizational transformation through reform, the 57 million leaked users have suffered irreparable damage. For today's enterprises, the choice is clear: either proactively build a trustworthy digital governance system or lose their hard-won market position in the next crisis. In the data-driven era, protecting trust is protecting the future.

10. Reference

1. Greenberg, A. (2017, November 21). *Hack brief: Uber paid off hackers to hide a 57-million user data breach*. WIRED.
<https://www.wired.com/story/uber-paid-off-hackers-to-hide-a-57-million-user-data-breach/>
2. Lee, B. D. (2017, November 22). *Uber concealed huge data breach*.
<https://www.bbc.com/news/technology-42075306>
3. Greenberg, A. (2017, November 22). Uber hid 57-Million user data breach for over a year. *WIRED*.
<https://www.wired.com/story/uber-paid-off-hackers-to-hide-a-57-million-user-data-breach/>
4. *Uber Settles FTC Allegations that It Made Deceptive Privacy and Data Security Claims*. (2017, March 31). Federal Trade Commission.
<https://www.ftc.gov/news-events/news/press-releases/2017/08/uber-settles-ftc-allegations-it-made-deceptive-privacy-data-security-claims>
5. Isaak, J., & Hanna, M. J. (2018). User data privacy: Facebook, Cambridge Analytica, and Privacy Protection. *Computer*, 51(8), 56–59. <https://doi.org/10.1109/mc.2018.3191268>
6. Martin, K. D., Borah, A., & Palmatier, R. W. (2016). Data privacy: Effects on customer and firm performance. *Journal of Marketing*, 81(1), 36–58.
<https://doi.org/10.1509/jm.15.0497>
7. Wong, J. C. (2017, November 27). Uber concealed massive hack that exposed data of 57m users and drivers. *The Guardian*.
<https://www.theguardian.com/technology/2017/nov/21/uber-data-hack-cyber-attack>
8. Sydney, I. (n.d.). *Chaos at Uber_The new CEO's challenge*. Scribd.
<https://www.scribd.com/document/623358613/Chaos-at-Uber-The-New-CEO-s-Challenge#:~:text=New%20CEO's%20Challenge-,The%20document%20discusses%20the%20challenges%20facing%20Uber's%20new%20CEO%20Data,integrity%20and%20earning%20customer%20trust.>

9. 1. Greenberg, A. (2017, November 21). Uber Paid Off Hackers to Hide a 57-Million User Data Breach. Wired.
<https://www.wired.com/story/uber-paid-off-hackers-to-hide-a-57-million-user-data-breach/>
10. Whittaker, Z. (2018, October 25). Two hackers behind 2016 Uber data breach have been indicted for another hack. TechCrunch; TechCrunch.
<https://techcrunch.com/2018/10/25/uber-hackers-indicted-lynda-breach/>
11. Lydia. (2019, May 13). Case study: Uber Technologies, Inc data breach. Medium; Golden Data.
<https://medium.com/golden-data/case-study-uber-technologies-inc-data-breach-7261484d6471>
12. Balakrishnan, A., & Bosa, D. (2017, November 21). Uber hid a hack that exposed data of 57 million users and drivers for more than a year. CNBC.
<https://www.cnbc.com/2017/11/21/uber-hack-exposes-data-of-57-million-users-and-drivers-report-says.html>
13. Federal Trade Commission. (2017, August 15). Uber Settles FTC Allegations that It Made Deceptive Privacy and Data Security Claims. Federal Trade Commission.
<https://www.ftc.gov/news-events/news/press-releases/2017/08/uber-settles-ftc-allegations-it-made-deceptive-privacy-data-security-claims>
14. Salmon, J. (2023, November 8). The Uber Breach Case Study: Cybersecurity Lessons Learned. HumanFirewall.
<https://humanfirewall.io/the-uber-breach-case-study-cybersecurity-lessons-learned/>
15. Bhuiyan, J., & Milmo, D. (2022, July 15). “Embrace the chaos”: a history of Uber’s rapid expansion and fall from favour. The Guardian.
<https://www.theguardian.com/news/2022/jul/15/embrace-the-chaos-a-history-of-ubers-rapid-expansion-and-fall-from-favour>
16. Mandarano, K. (2023, April 4). *When in doubt, report it out: What Uber’s ex-CSO’s conviction means for companies’ mandatory reporting duties following a data breach.* University of Maryland Francis King Carey School of Law.
<https://www.law.umaryland.edu/content/articles/name-659560-en.html>

17. United States Attorney's Office, Northern District of California. (2022, October 5). *Former chief security officer of Uber convicted of federal charges for covering up data breach involving millions of Uber user records*. U.S. Department of Justice. <https://www.justice.gov/usao-ndca/pr/former-chief-security-officer-uber-convicted-federal-charges-covering-data-breach-involving>
18. *Former chief security officer of Uber convicted of federal charges for*. (2022, October 6). <https://www.justice.gov/usao-ndca/pr/former-chief-security-officer-uber-convicted-federal-charges-covering-data-breach>
19. Newman, L. H. (2022, October 7). *The Uber data breach conviction shows security execs what not to do*. WIRED. <https://www.wired.com/story/uber-joe-sullivan-conviction/>
20. Greenberg, A. (2017, November 21). *Hack briefing: Uber paid off hackers to hide a 57-million user data breach*. WIRED. <https://www.wired.com/story/uber-paid-off-hackers-to-hide-a-57-million-user-data-breach/>
21. Dwyer, K. (2022, October 19). *The fallout from the first trial of a corporate executive for 'covering up' a data breach*. Lawfare. <https://www.lawfaremedia.org/article/fallout-first-trial-corporate-executive-covering-data-breach>
22. HackerOne. (July, 2019). *Vulnerability disclosure guidelines*. <https://www.hackerone.com/terms/disclosure-guidelines>
23. Cook, J., Patel, R., & O'Rourke, J. S. (2018). *Uber Technologies, Inc.: Accountability in corporate culture*. Eugene D. Fanning Center for Business Communication, Mendoza College of Business, University of Notre Dame. https://www.researchgate.net/publication/332531033_Uber_Technologies_Inc_Accountability_in_Corporate_Culture

24. Association for Computing Machinery. (2020). ACM Code of Ethics and Professional Conduct. <https://www.acm.org/code-of-ethics>
25. Rueter, S. (2025, July 5). Understanding Deontology: Ethics and Principles. Philosophos. <https://www.philosophos.org/ethics-deontology>
26. Isaac, M., & Frenkel, S. (2017, November 21). Uber Hid 2016 Breach, Paying Hackers to Delete Stolen Data. The New York Times. <https://www.nytimes.com/2017/11/21/technology/uber-hack.html>
27. Mill, J. S. (1863/2001). Utilitarianism. Hackett Publishing Company. <https://books.google.com.my/books?id=ilq0AwAAQBAJ&lpg=PT3&ots=3hbC-nub75&dq=Utilitarianism%20by%20John%20Stuart%20Mill&lr&pg=PT3#v=onepage&q=Utilitarianism%20by%20John%20Stuart%20Mill&f=false>
28. 2. Rueter, S. (2023, June 22). Understanding utilitarianism: A guide. Philosophos. <https://www.philosophos.org/ethics-utilitarianism>
29. 3. Shaw, W. H. (2016). Business ethics: A textbook with cases (9th ed.). Cengage Learning. https://www.google.com.my/books/edition/Business_Ethics/vM2a0AEACAAJ?hl=en
30. Aristotle. (trans. Hursthouse, R., & Pettigrove, G.). (2018). Virtue ethics. In The Stanford Encyclopedia of Philosophy (Fall 2018 ed.). <https://plato.stanford.edu/entries/ethics-virtue/>
31. Hartman, L. P., DesJardins, J., & MacDonald, C. (2020). Business ethics: Decision-making for personal integrity and social responsibility (5th ed.). McGraw-Hill Education. <https://www.mheducation.com/highered/product/Business-Ethics-Decision-Making-for-Personal-Integrity-and-Social-Responsibility-Hartman.html>
32. Rueter, S. (2024, April 14). Understanding virtue ethics: Building character and moral strength. Philosophos. <https://www.philosophos.org/ethics-virtue-ethics>
33. California Consumer Privacy Act (CCPA), Cal. Civ. Code §§ 1798.100-1798.199 (2018). https://leginfo.ca.gov/faces/codes_displayText.xhtml?division=3.&part=4.&lawCode=CIV&title=1.81.5
34. de la Torre, L. F. (2019, May 13). The Uber breach story: On how security woes can lead to a criminal complaint. Medium.

<https://medium.com/golden-data/case-study-uber-technologies-inc-data-breach-7261484d6471>

35. Dwyer, D. (2022, October 6). The fallout from the first trial of a corporate executive for covering up a data breach. Lawfare. <https://www.lawfaremedia.org/article/fallout-first-trial-corporate-executive-covering-data-breach>
36. European Parliament and Council of the European Union. (2016). Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation). Official Journal of the European Union, L119, 1-88. <https://eur-lex.europa.eu/eli/reg/2016/679/oj>
37. National Institute of Standards and Technology. (2018). Framework for improving critical infrastructure cybersecurity, Version 1.1. U.S. Department of Commerce. <https://doi.org/10.6028/NIST.CSWP.04162018>
38. United Nations Office on Drugs and Crime. (2001). Convention on Cybercrime (Budapest Convention), CETS No. 185. Council of Europe. <https://www.europarl.europa.eu/cmsdata/179163/20090225ATT50418EN.pdf>
39. Lee, B. D. (2017, November 22). Uber concealed huge data breach. <https://www.bbc.com/news/technology-42075306>
40. User data privacy: Facebook, Cambridge Analytica, and Privacy Protection. (2018, August 1). IEEE Journals & Magazine | IEEE Xplore. <https://ieeexplore.ieee.org/document/8436400>
41. Wong, J. C. (2017, November 27). Uber concealed massive hack that exposed data of 57m users and drivers. The Guardian. <https://www.theguardian.com/technology/2017/nov/21/uber-data-hack-cyber-attack>
42. Uber Settles FTC Allegations that It Made Deceptive Privacy and Data Security Claims. (2025, March 31). Federal Trade Commission. <https://www.ftc.gov/news-events/news/press-releases/2017/08/uber-settles-ftc-allegations-it-made-deceptive-privacy-data-security-claims>
43. Greenberg, A. (2017, November 22). Uber hid 57-Million user data breach for over a year. WIRED.

<https://www.wired.com/story/uber-paid-off-hackers-to-hide-a-57-million-user-data-breach/>

44. Kaplan, C. (2025, August 1). Former Uber CSO gets a free ride – to prison. HALOCK. <https://www.halock.com/former-uber-cso-gets-a-free-ride-to-prison/>
45. Martin, K. D., Borah, A., & Palmatier, R. W. (2016). Data Privacy: Effects on Customer and Firm Performance. *Journal of Marketing*, 81(1), 36–58. <https://doi.org/10.1509/jm.15.0497>
46. Dara Khosrowshahi. (2017, November 21). 2016 data security incident. Uber Newsroom. <https://www.uber.com/en-CH/newsroom/2016-data-incident/>
47. Dara Khosrowshahi. (2017, November 21). 2016 Data Security Incident. Uber Newsroom. <https://www.uber.com/en-CH/newsroom/2016-data-incident/>
48. Furneaux, A. (2016). What to Learn From Uber’s 2016 Data Breach... Cybersaint.io. <https://www.cybersaint.io/blog/uber-hack-undisclosed>
49. Lee, D. (2017, November 22). Uber concealed huge data breach. BBC News. <https://www.bbc.com/news/technology-42075306>
50. Salmon, J. (2023, November 8). The Uber Breach Case Study: Cybersecurity Lessons Learned. HumanFirewall. <https://humanfirewall.io/the-uber-breach-case-study-cybersecurity-lessons-learned/>
51. National Institute of Standards and Technology. (2020, August 11). Zero trust architecture: NIST publishes SP 800-207. U.S. Department of Commerce. <https://www.nist.gov/news-events/news/2020/08/zero-trust-architecture-nist-publishes-sp-800-207>
52. Wong, J. C. (2017, November 27). Uber concealed massive hack that exposed data of 57m users and drivers. The Guardian; The Guardian. <https://www.theguardian.com/technology/2017/nov/21/uber-data-hack-cyber-attack>
53. Federal Trade Commission . (2018, April 11). Uber Agrees to Expanded Settlement with FTC Related to Privacy, Security Claims. Federal Trade Commission. <https://www.ftc.gov/news-events/news/press-releases/2018/04/uber-agrees-expanded-settlement-ftc-related-privacy-security-claims>

54. Furneaux, A. (2018, January 4). What to Learn From Uber's 2016 Data Breach... That The Company Just Disclosed This Month. Wwww.cybersaint.io.
<https://www.cybersaint.io/blog/uber-hack-undisclosed>
55. Lohrmann, D. (2017). After Uber Data Breach: Lessons for All of Us. Govtech.com.
<https://www.govtech.com/blogs/lohrmann-on-cybersecurity/after-uber-data-breach-lessons-for-all-of-us.html>
56. European Parliament and Council of the European Union. (2016). Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation). Official Journal of the European Union, L119, 1–88. <https://eur-lex.europa.eu/eli/reg/2016/679/oj>
57. Federal Trade Commission. (2018, October 26). Federal Trade Commission gives final approval to settlement with Uber.
<https://www.ftc.gov/news-events/news/press-releases/2018/10/federal-trade-commission-gives-final-approval-settlement-uber>